

- | | | | |
|--|---|---|---|
| 10. A process having multiple threads of control implies | 1 | 2 | 3 |
| (A) it can do only one task at a time, but much faster | | | |
| (B) it can do only one task at a time & very slow | | | |
| (C) it can do more than one task at a time | | | |
| (D) it has to use only one thread per process | | | |
| 11. A Computer has 6 tape drives, which n processes competing for them. Each process may need two drives. What is the Maximum value of n for the system to be deadlock free? | 1 | 3 | 3 |
| (A) 6 | | | |
| (B) 5 | | | |
| (C) 4 | | | |
| (D) 3 | | | |
| 12. If we preempt a resource from a process, the process cannot continue with its normal execution and it must be _____ | 1 | 2 | 3 |
| (A) aborted | | | |
| (B) rolled back | | | |
| (C) terminated | | | |
| (D) queued | | | |
| 13. Which of the page replacement policy sometimes leads to more page faults when size of memory is increased? | 1 | 2 | 4 |
| (A) LRU | | | |
| (B) MRU | | | |
| (C) FIFO | | | |
| (D) Optimal | | | |
| 14. Thrashing implies | 1 | 2 | 4 |
| (A) Reduces Page I/O | | | |
| (B) Decrease the degree of Multi Programming | | | |
| (C) Excessive Page I/O | | | |
| (D) Improve the system performance | | | |
| 15. The swap space in the disk is used for | 1 | 1 | 4 |
| (A) Saving temporary html pages | | | |
| (B) Saving process data | | | |
| (C) Storing the super-block | | | |
| (D) Storing device drivers | | | |
| 16. The time taken to move the disk arm to the desired cylinder is called the _____ | 1 | 1 | 4 |
| (A) Rotational Latency | | | |
| (B) Seek Time | | | |
| (C) Random Access Time | | | |
| (D) Positioning Time | | | |
| 17. The access matrix represent | 1 | 1 | 5 |
| (A) Rows-Domains, Columns-Objects | | | |
| (B) Rows-Objects, Columns-Domains | | | |
| (C) Rows-Access List, Columns-Domains | | | |
| (D) Rows-Domains, Columns-Access list | | | |
| 18. Which of the following objects require protection? | 1 | 2 | 5 |
| (A) Monitor | | | |
| (B) CPU | | | |
| (C) Key Board | | | |
| (D) Memory | | | |
| 19. The FALSE Statement about encryption is | 1 | 1 | 5 |
| (A) It is used to protect data from unauthorized access during transmission | | | |
| (B) It is used to ensure user authentication | | | |
| (C) It is used to ensure data integrity | | | |
| (D) It is used to ensure data corruption doesn't happens | | | |
| 20. Which is not a common file permission? | 1 | 1 | 5 |
| (A) read | | | |
| (B) write | | | |
| (C) stop | | | |
| (D) execute | | | |

PART - B (4 × 10 = 40 Marks)

Answer **any** 4 Questions

Marks BL .CO

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|---|----|---|---|
| 21. Describe the Services provided by Operating Systems | 10 | 2 | 1 |
|---|----|---|---|

22. Illustrate readers writers classical synchronization problem & write the Pseudo code with proper explanation 10 3 2

23. 10 4 3

Process ID	Burst Time	Priority
P1	8	4
P2	6	1
P3	1	2
P4	9	2
P5	3	3

Consider the above set of process at ready queue for it execution. The process are assumed arrived in the order P1,P2,P3,P4,P5 at time 0

1. Draw the Gantt chart to illustrate the execution of the process using FCFS,SJF, Priority, Round Robin
2. Find the Turn around time and Waiting time of each process for each scheduling algorithm
3. Compare which Scheduling algorithm gives best result

24. Consider the following page-reference string: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults would occur for the following replacement algorithms, assuming four, five frames? Remember that all frames are initially empty, so your first unique pages will all cost one fault each. 10 4 4

1. FIFO replacement
 2. LRU replacement
 3. Optimal replacement
- Compare your results.

25. Explain the Various Directory and Disk Structures 10 4 4
26. Explain the common methods by which programs cause security breaches 10 3 5

PART - C (1 × 15 = 15 Marks)

Answer any 1 Questions

27. 15 4 3

	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

Solve the above problem using banker's algorithm & find

- a. NEED Matrix
- b. safe sequence
- c. If a request from process P1 arrives for (0,2,2,0), can the request be granted immediately?

28. In Disk Scheduling, read-write head initially at 100. Disk contains 200 tracks. The I/O request is arriving in the following order 55, 58, 39, 18, 90, 160, 150, 38, 184 Calculate the average seek length using FIFO,SSTF, SCAN, C-SCAN algorithm 15 4 4
